

Tyler Forstrom

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EDUCATION

Cornell University, College of Engineering

Bachelor of Science in Computer Science, Minor in Business

Ithaca, NY

Expected December 2026

EXPERIENCE

App Developer Intern, Kahua (Construction Software Company)

Internship

- Developed web applications to support construction project workflows and improve data accessibility.
- Built and integrated full-stack features with a focus on performance and user-friendly design.
- Collaborated on tools that streamline communication and organization for project teams.

Adaptive Ecosystem Lab — Multi-Agent Reinforcement Learning (PyTorch)

Personal Project | 2026

github.com/TylerJForstrom/EvolutionSim

- Built a vectorized 2D ecosystem and a multi-agent reinforcement learning trainer from scratch in NumPy and PyTorch, where five species (herbivores, predators, decomposers, pollinators, engineers) each learn their own policy network concurrently from reward alone.
- Implemented Proximal Policy Optimization (PPO) from scratch and diagnosed three training-collapse failure modes (policy mode-collapse, food-web NaNs, ~70% wasted compute).
- Fixed instability with a PPO ratio-overflow guard, a global carrying-capacity cap, and an adaptive entropy controller, taking runs from always-collapsing to reliably stable.

BanditFeed — Contextual-Bandit Recommendation System (FastAPI)

Personal Project | 2025-2026

github.com/TylerJForstrom/BanditFeed

- Built an interpretable article recommender that ranks real RSS content with contextual bandits, balancing explore-vs-exploit and updating its beliefs online from each click or skip.
- Surfaced a per-article explanation of why each item was ranked, making the model's reward estimates and uncertainty visible in real time.

Agent-Based Stock Market Simulator & Strategy Validation Pipeline

Personal Project | 2025-2026

github.com/TylerJForstrom/Stock-Market-Sim

- Built an agent-based limit-order-book market simulator in pure Python where prices emerge from competing trading agents (market makers, momentum, mean-reversion, and institutional order flow).
- Designed a statistical validation pipeline (Deflated Sharpe Ratio with multiple-testing correction, walk-forward out-of-sample testing, and realistic trading costs) to test strategies honestly and avoid false positives.
- Found no reliable directional edge across eight strategy families, and built a volatility forecaster competitive with industry-standard models (HAR-RV, EWMA) out-of-sample on equities and crypto.
- Wrote ~350 automated tests with continuous integration and a tamper-evident research log; implemented all statistics from scratch with no external numerical libraries.

Sketch-to-Image Enhancement Using Deep Learning (PyTorch)

Personal Project | Hartford, CT

- Built a sketch-to-image generation model using custom U-Net architectures and GAN training (PatchGAN discriminator) in PyTorch.
- Implemented perceptual, edge, and L1 reconstruction losses to improve realism and stability, with a multi-stage training curriculum (warmup, then GAN, then deep-model refinement).
- Developed a Streamlit app enabling users to draw or upload sketches and generate colorized outputs in real time.

ACTIVITIES

Investments in Technology @ Cornell

Ithaca, NY | Jan 2023

Founder, Treasurer & Portfolio Manager

- Co-founded a club focused on teaching students about investing in the tech industry and managing a real investment portfolio.
- Collaborated with a team to develop educational materials, organize meetings, and lead discussions on investment strategies, market trends, and risk management.

ADDITIONAL INFORMATION

Technical Skills: Python, Java, C, OCaml, SQL · PyTorch, TensorFlow, NumPy · Reinforcement learning (PPO, multi-agent), GANs, contextual bandits, NLP sentiment, applied statistics

Interests: Martial Arts, Gaming, Investing, Weightlifting, Golf